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Configurable quantum-computing control system

Abstract

A system for controlling a quantum processor includes a system controller having a plurality of input channels and output channels connected to qubit readout instruments and qubit control instruments. The system controller is implemented at least in part as an FPGA. It forms a register bank for receiving the data from the input channels, a feedback block for processing the data from the register bank, and an output interface for sending the results of the feedback block to the control instruments. The feedback block can be programmed by an FPGA configuration device. The FPGA configuration device is able to process various types of feedback block definitions, which e.g. describe lookup tables, minimizers, Boolean operations, etc. A configuration device includes a system compiler agnostic of the specific implementation of the feedback block for controlling the experiments and the operation of the instruments.

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Background/Summary

TECHNICAL FIELD

(1) The invention relates to a system for controlling a quantum processor.

BACKGROUND ART

(2) A system for controlling a cryogenic quantum processor typically comprises a system controller having a plurality of input channels and output channels, a plurality of qubit readout instruments connected to the input channels, and a plurality of qubit control instruments connected to the output channels.

(3) The qubit control instruments are adapted to send signals into the quantum processor to set up or adjust its state while the qubit readout instruments probe the quantum processor and read out one or more of the qubits at certain times, e.g. for result sampling or error correction.

(4) The system controller provides a feedback mechanism controlling the control instruments in response to the signals measured by the readout instruments. It must therefore be fast and deterministic. Further, in order to adapt it to the requirements of the specific quantum processor and its desired method of operation, the system controller should be configurable by the user.

(5) A system of this type is e.g. described by C. A. Ryan et al. “Hardware for Dynamic Quantum Computing”, arXiv:1704.08314v1, 28 Apr. 2017.

(6) The feedback implemented by the system controller can e.g. be used for state initialization as described in the reference above and/or for Quantum Error Correction (QEC) as e.g. described e.g. by P. Das et al. “A Scalable Decoder Micro-architecture for Fault-Tolerant Quantum Computing”, arXiv:2001.06598v1, 18 Jan. 2020.

DISCLOSURE OF THE INVENTION

(7) The problem to be solved by the present invention is to provide a system of the type above that allows the experimenter to set up the system controller in a versatile yet simple manner.

(8) This problem is solved by the system of claim 1.

(9) Accordingly, the system comprises at least the following elements: A system controller having a plurality of input channels and output channels: This is the part of the device that implements the feedback logic between the measured signals from the quantum processor and the signals to be sent to the quantum processor. Each of the input channels carries address data and measurement data, whose roles are described below. A plurality of qubit readout instruments connected to the input channels of the system controller: The readout instruments comprise the circuitry to read one single qubit or combinations of qubits from the quantum processor and to convert them into measurement data. In addition, they may comprise storage for address data to be sent along with the measurement data through their respective input channel to let the system controller know where the measurement data is to be stored. A plurality of qubit control instruments connected to the

output channels: The control instruments comprise circuitry to control the qubits of the quantum processor.

(10) The system controller comprises at least one FPGA, i.e. a programmable gate array. This one or more FPGA form(s) at least the following parts of the system controller:

(11) a) An input interface: The input interface comprises address decoder circuitry connected to the input channels.

(12) b) A register bank: The register bank comprises a plurality of registers connected to the input interface.

(13) c) A programmable feedback block: The feedback block has inputs connected to the register bank, and it has outputs. It implements the feedback logic of the system controller.

(14) d) An output interface: The output interface connects the outputs of the feedback block to the output channels.

(15) The input interface is adapted to send the measurement data of the input channel to one of the registers of the register bank as specified by the address data in a given input channel. In other words, the readout instruments are able to specify the register(s) into which the measurement data is to be stored.

(16) The system further comprises an FPGA configuration device connected to at least one FPGA. The FPGA configuration device is adapted to carry out at least the following functions: Receiving at least one feedback block definition: This is a definition of the functions to be carried out by the feedback block. This feedback block definition may e.g. comprise a textual configuration description (e.g. in VHDL) defining the logical configuration of at least part of the feedback block, a bit stream configuration to be fed directly to the FPGA, and/or a configuration dataset to be stored into registers formed by the configured FPGA. Programming a section of the FPGA to form said feedback block in accordance with the feedback block definition. In other words, the FPGA configuration device translates, where necessary, the feedback block definition and feeds it to the FPGA. For example, it generates a bit stream from a textual logic definition and feeds it to the FPGA, and/or it forwards a received bit stream to the FPGA, and/or it feeds a configuration dataset into registers formed by the configured FPGA.

(17) Note: the number of readout instruments, control instruments, and registers are typically different from each other.

(18) Advantageously, the FPGA configuration device has several “modes” for programming said section of the FPGA. The different modes are adapted to program the feedback block to implement different algorithms for connecting the registers to the outputs of feedback block, and they can be optimized to make it easy for the user to specify the elements or parameters of the algorithm(s) to be used.

(19) In particular, the FPGA configuration device may have a “LUT-mode”. When in the LUT mode, the FPGA configuration device is adapted to program the section of the FPGA to form a lookup table. This lookup table has a selection input connected to the register bank and a data output connected to the output interface. It can be used to implement the feedback as a simple lookup table that provides the values (or parts thereof) to be sent to some or all of the control instruments as a function of the values in some or all of the registers.

(20) The FPGA configuration device may also have an “ML-mode”. When in the ML mode, the FPGA configuration device is adapted to program the section of the FPGA to implement a machine learning model. This model may advantageously implement an artificial neural network.

Alternatively, or in addition thereto, it may implement at least one of a decision tree, a support vector machine, a regression analyzer, a Bayesian network, or a genetic algorithm.

(21) The FPGA configuration device may also have an “FPGA direct programming mode”. When in this mode, the FPGA configuration device is adapted to receive gate-level instructions for programming the section of the FPGA. Advantageously, the FPGA configuration device is adapted to receive a set of VHDL instructions and to use these instructions for programming the FPGA in

said section.

(22) In another embodiment, the system may further comprise a co-processor, such as a GPU or other hardware optimized for numerical processing as e.g. described in more detail below.

(23) The coprocessor is separate from the FPGA circuitry of the system controller and advantageously not an FPGA. It is connected to the FPGA. In this case, the FPGA configuration device may also have a “coprocessor programming mode”. When in this mode, the FPGA configuration device is adapted to program the section of the FPGA to exchange data with the coprocessor in order to execute its calculations, at least in part, on the coprocessor.

(24) The FPGA configuration device may also have one or more other modes as specified below.

(25) In one embodiment, the FPGA configuration device may be adapted to carry out the following steps: Merge the “feedback block definition” with information descriptive of the input interface, the register bank, and the output interface in order to generate FPGA configuration data. Such FPGA configuration data may e.g. be a series of VHDL instructions and/or a series of configuration values to be sent into the configuration registers of the FPGA. Program the at least one FPGA according to said FPGA configuration data in order to form the input interface, register bank, programmable feedback block, and output interface in said at least one FPGA.

(26) This allows to greatly reduce the complexity of configuring the FPGA because the programming of its standard parts, i.e. of the input interface, register bank, programmable feedback block, and output interface, are taken care of by the FPGA configuration device. The user only needs to concentrate on the part that controls the quantum processor's functionality, i.e. on the feedback block.

(27) In another important embodiment, the system comprises a system configuration device connected to the readout instruments and to the control instruments. The system configuration device is adapted to program the readout instruments and the control instruments. Advantageously, the system configuration device is a device separate from the FPGA configuration device.

(28) Advantageously, each readout instrument comprises at least one address memory storing the address to be transmitted as address data together with the measurement data on the input channel connecting the readout instrument to the system controller. This address memory may e.g. be part of a program memory storing instructions to be executed by the readout instrument.

(29) The system configuration device may be adapted to write the address into the address memory. This allows the system configuration device to define the register(s) a readout instrument is to write into.

(30) The system configuration device may comprise a delay input connected to the FPGA configuration device for receiving a value descriptive of the time delay between the input channels and the output channels in the current configuration of the system controller. The system configuration device may e.g. use this time delay for synchronizing the operation of the control instruments and the readout instruments.

(31) In that case, the FPGA configuration device is adapted to calculate the time delay as a function of the feedback control definition, i.e. it analyses the feedback control definition and calculates the time delay therefrom. In a general case, the FPGA configuration device is adapted to process at least two possible feedback control definitions that result in two different time delays.

(32) In a further embodiment, each register not only comprises a “register data output” carrying a signal indicative of the data stored in the register, but it also comprises a “new value output”. The new value output carries a signal indicative of a write event into the register, i.e. the new value output changes state when one of the readout instruments has issued a write command into the respective register by transmitting the register's address through the address data in its input channel.

(33) Hence, the new value output changes state and becomes active even if the actual value in the register does not change.

(34) Both, the register data output as well as the new value output of the registers, are connected to

the feedback block. This allows the feedback block to recognize the time when a recalculation of its outputs may be required.

(35) In particular, the feedback block may be adapted to initiate a recalculation of at least some of its outputs if at least some of the new value outputs flags a write operation in its register.

(36) Advantageously, the feedback block may comprise a sensitivity list storing a sensitivity information indicative of a subset of the registers and be adapted to initiate a recalculation if the new value output of at least one register of the subset flags a write operation in its register. In this case, advantageously, the FPGA configuration device may be adapted to set this sensitivity information. The sensitivity information may be a function of the feedback block definition, i.e. the feedback block definition forms an explicit or implicit declaration of the registers the feedback block (or part thereof) depends on.

(37) The invention also relates to the use of such a system for controlling a quantum processor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings, wherein:

(2) FIG. 1 shows a general overview of an embodiment of the system,

(3) FIG. 2 is a more detailed view of an embodiment of the system controller,

(4) FIG. 3 is a more detailed view of a FPGA configuration device,

(5) FIG. 4 shows an example of a lookup table,

(6) FIG. 5 is a more detailed view of an embodiment of a readout instrument, and

(7) FIG. 6 is a more detailed view of an embodiment of a control instrument.

MODES FOR CARRYING OUT THE INVENTION

(8) Overview

(9) FIG. 1 shows an overview of an embodiment of a system for controlling a quantum processor **10**.

(10) Quantum processor **10** may be implemented in any suitable quantum processor technology. It has a first plurality of interacting qubits, a second plurality of inputs **12** for setting the states of the qubits, and a third plurality of outputs **14** for detecting the states. Typically, the first, second, and third pluralities will differ in numbers.

(11) The system **16** has the purpose to set the qubits, to sample the qubits, and/or to control the qubits, e.g. for setting up the state of the qubits, for reading results from quantum processor **10**, or for error correction in QEC scheme as mentioned above.

(12) System **16** comprises a system controller **18** having a plurality of input channels **20** and a plurality of output channels **22**. FIG. 1 shows four of each of these channels, but that number will, in most applications, typically be larger, and the numbers of input channels and output channels will typically differ.

(13) The outputs **14** of quantum processor **10** are connected to qubit readout instruments **24**, and each readout instrument **24** is connected to system controller **18** by means of at least one of the input channels **20**.

(14) Typically, each readout instrument **24** generates pulses to be fed to quantum processor **10** (not shown in the figures) and then measures the response at the outputs **14**.

(15) The output channels **22** of system controller **18** are connected to qubit control instruments **26**, and each control instrument is connected to at least one of the inputs **12** of quantum processor **10**.

(16) Again, there may be any suitable numbers of readout instruments **24** and the control instruments **26**.

- (17) The system further comprises an FPGA configuration device **28** for configuring a user-configurable section of system controller **18**.
- (18) It may also comprise a system configuration device **30** connected to the readout instruments **24**, the control instruments **26**, and/or the system controller **18** and adapted to program them as described in more detail below.
- (19) The system may further e.g. comprise a first higher-level control device **32** for controlling FPGA configuration device **28** and/or a second higher-level control device **34** for controlling system configuration device **30**.
- (20) In operation, system controller **18** controls and synchronizes the control instruments **26** for setting up and/or correcting the qubit states of quantum processor and the readout instruments **24** for obtaining signals. FPGA control device **28** controls the setup of the feedback logics between the input channels **20** and the output channels **22**. System configuration device **30** controls the operating parameters of the readout instruments **24** and the control instruments **26** as well as the process control incorporated in system controller **18**.
- (21) Possible designs and functionalities of the individual components will be described in more detail in the following sections.
- (22) System Controller
- (23) FIG. 2 shows a more detailed view of an embodiment of system controller **18**.
- (24) Typically, system controller **18** is implemented, at least in part, as an FPGA **19**, for the high speed, high degree of parallelism, and the well-defined delay time of such FPGA circuitry.
- (25) In particular, it comprises a communication and synchronization section **40**, which forms the interface for the input channels **20** and the output channels **22**. In particular, it serializes and deserializes data, it feeds a common, global clock signal to each channel **22**, **20**, and it handles synchronization signals at each channel **22**, **20**.
- (26) Communication and synchronization section **40** may also comprise an experiment controller **41** to control a series of experiments. Its role will be described below.
- (27) As will be described below, the input channels **20** carry address data to and measurement data. Typically, the address data is a digital data value indicative of the address of one of several registers, as described below, and the measurement data is a binary value describing the result of a measurement.
- (28) The address data and measurement data are fed to an input interface **42**, which controls a register bank **44**.
- (29) Input interface **42** comprises address decoder circuitry **46** for decoding the address data in order access the appropriate register **48** in register bank **44**. The digital data is then stored in the register **48** specified by the address data.
- (30) As above, even though FIG. 2 shows eight registers **48**, this is for illustration purposes only. The actual number of registers in a device may be different and will typical be larger. Typically, there is a least one register per qubit of quantum processor **10**. Typically, there are at least 7 registers, in particular at least 100 registers.
- (31) Each register **48** has two outputs: A “data output” **50** carries a signal indicative of the value stored in the register, e.g. a logical 1 if the register stores a 1-value and a logical 0 if the register stores a 0-value. A “new value output” **52** carries a signal indicative of a write event into the register. A “write event” is a write operation generated by an input channel **20** sending data to the respective register even if the actual value of the register is not changed by the operation. In other words, each new value output **52** flags the occurrence of a write operation in its register **48**. The new value output may e.g. be a binary signal set into an “on” state upon a write operation and reset e.g. after the feedback block (to be described below) has calculated its output values. Alternatively, and typically, the new value output may be reset after a predefined time interval, typically one clock cycle.
- (32) Register bank **44**, i.e. the data outputs **50** and the new value outputs **52**, is connected to a

programmable feedback block **54**.

(33) In the shown embodiment, feedback block **54** is embodied as FPGA circuitry and implements the algorithm used to generate the values for the output channels **22** from the values in register bank **44**. It will be described in more detail below. But in one embodiment, it may e.g. comprise a lookup table **56** as shown.

(34) Feedback block **54** comprises inputs **57** connected to register bank **44**.

(35) In addition, feedback block **54** may comprise a sensitivity list **58** storing a sensitivity information. This sensitivity information may e.g. be a series of bits, with each bit corresponding to one of the registers **48**, and/or it may be a list of register addresses.

(36) If, for example, the new value output **52** of at least one register **48** flags that a write operation has occurred in the register and the corresponding bit in sensitivity list is 1 (or the register address is in the list of register addresses), feedback block **54** will recalculate the values at its outputs **60** from the values at its inputs **57**.

(37) In another example, feedback block **54** will recalculate the values at its output **60** only if the new value output **52** of all the registers **48** that are specified in the sensitivity list **58** flag a change.

(38) In yet another embodiment, the sensitivity list may specify yet another logic combination of the new value outputs **52** that is required to trigger a recalculation.

(39) There may be several such sensitivity lists, each for a different purpose. For example, one for the initialization phase and second one for the QEC phase.

(40) In more general terms, the sensitivity information specifies a “subset” of the registers **48**, and feedback block **54** is adapted to initiate a recalculation, if a given combination (such as an “and” combination or an “or” combination or yet another type of logical combination) of this subset flags a write operation in the respective register(s).

(41) The outputs **60** of feedback block **54** (which also form the inputs of output interface **61**) are connected to an output interface **61**, where they are converted to signals to be fed via communication and synchronization section **40** to the output channels **22**.

(42) Advantageously, each output **60** is a one-bit or multibit value to be transmitted to a specific control instrument **26**. Output interface **61** is adapted to send each one of these values to their respective control instrument **26** via one of the output channels **22**.

(43) System controller **18** may further comprise a JTAG interface **64** for low-level access to its FPGA circuitry **19** and a control interface **65** for being connected to FPGA configuration device **28**.

(44) System controller **18** may also comprise an interface **66** for connecting feedback block **54** to external circuitry, such as a coprocessor **67**. Interface **66** may e.g. be a PCI or AXI bus.

(45) Coprocessor **67** comprises optimized hardware for fast numerical processing and may e.g. be a GPU or a processor dedicated to other types of processing, such as a Versal™ coprocessor by Xilinx Inc. This physical coprocessor may be implemented in the same chip as the FPGA, on the same PCB as the FPGA, or on another PCB.

(46) More details about these components are provided below.

(47) FPGA Configuration Device

(48) FPGA configuration device **28** is typically a computer device running software that is configured to receive a “feedback block definition” and to program a section of the FPGA **19** of system controller **18** accordingly.

(49) The section it programs typically corresponds to feedback block **54** or comprises feedback block **54**.

(50) FIG. 3 shows a block diagram of an embodiment of FPGA configuration device **28**. It typically supports different types of feedback block definitions **68**, which are processed by suitable processing modules **69** to be converted into configuration information to be fed through an interface unit **70** to control interface **65** of system controller **18**.

(51) In the context of the present text, the different types of feedback block definitions **68** and their respective processing modules **70** are designated as “modes” of configuration device **28**.

(52) Advantageously, configuration device **28** is adapted to provide different modes, with the different modes being adapted to program different algorithms into feedback block **54** for connecting the registers **48** to the outputs **60** of feedback block **54**. Depending on the task to be achieved, the user can employ one or more of these modes for implementing feedback block **54**.
(53) In some embodiments, the feedback block definitions **68** may e.g. comprise at least one configuration dataset that is directly forwarded into registers configured in feedback block **54** for setting values therein.

(54) In the following, some possible modes are described.

(55) LUT-Mode:

(56) Configuration device **28** may comprise an “LUT-mode”. In this mode, it is adapted to program feedback block **54** to form a lookup table (LUT) storing a plurality of coefficients. The lookup table comprises a selection input connected to register bank **44** (i.e. to the data outputs **50** of the registers **48**) and a result output connected to output interface **61**.

(57) An example for such a lookup table **56** is shown in FIG. **4**. It comprises a plurality of storage cells **72**, with each storage cell **72** storing a multibit value (a “coefficient”). Selection input **74** of lookup table **56** connects the data outputs **50** of at least a subset of the registers **48** to an address decoder **76**, optionally via a switch **75**. Switch **75** may be provided to select a subset of the registers **48** to be connected to address decoder **76**. Address decoder **76** interprets the bit pattern at selection input **74** or as received from switch **75**, i.e. at least a subset of the values from the registers, as an address for selecting a single one of the storage cells **72**. The value in that storage cell is then fed via an output section **78** to a result output **80**, which is connected to output interface **61**.

(58) Output section **78** and/or output interface **61** define which bits of the registers **48** are fed to which control instruments **26**.

(59) In the LUT-mode, block definition **68** may e.g. comprise a list of the registers **48** to be connected to the table, a list of the output channels **22** to be connected to the table, and the coefficients to be stored in the storage cells **72**. As mentioned, block definition **68** may also comprise a configuration dataset to be directly forwarded into the storage cells **72**.

(60) The values of the coefficients may be pre-computed on a classical computer.

(61) ML-Mode:

(62) Configuration device **28** may comprise an “ML-mode”. In this mode, it is adapted to program feedback block **54** as a machine learning model, e.g. in cooperation with coprocessor **67**. Various suitable models of this type are known to the skilled person, in particular artificial neural networks. Other models, though, may be used as well, in particular support vector machines, and/or also decision trees, regression analyzers, Bayesian networks, or genetic algorithms.

(63) The model can be implemented in the circuitry of the FPGA **19** and/or in external circuitry, such as coprocessor **67**.

(64) In ML-mode, block definition **68** may e.g. comprise a specification of the type of model and its configuration parameters as well as lists of the registers **48** and of the output channels **22** to be connected to it.

(65) The model may be trained on a classical computer. For example, for a particular quantum processor with its specific error map, a minimization algorithm may be approximated that determines the most likely underlying error based on readout of ancilla qubits.

(66) FPGA Direct Programming Mode:

(67) Configuration device **28** may comprise an “FPGA direct programming mode”. In this mode, block definition **68** comprises a gate-level FPGA instructions for programming the section of the FPGA **19** corresponding to feedback block **54**. For example, block definition **68** may comprise VHDL instructions for defining the configuration of the FPGA circuitry of feedback block **56**.

(68) In this programming mode, the user may e.g. specify multipliers, adders, memory cells, logic elements, multiply-accumulators, DSP slices, switches and/or multiplexers to be implemented in

feedback block **54**.

(69) Coprocessor Programming Mode:

(70) Configuration device **28** may comprise a “coprocessor programming mode”. In this mode, it is adapted to program feedback block **54** to send data derived from at least part of the registers **48** to coprocessor **67** for processing and to receive at least some of the results back from coprocessor **67** and send them to output interface **61**.

(71) In this mode, block definition **68** may e.g. comprise information indicative of the register values to be forwarded to coprocessor **67** and how to dispatch the results returned from coprocessor **67** to output interface **61**. Block definition **68** may also comprise code to be executed by coprocessor **67**—alternatively, said code may be sent through different means to coprocessor **67**.

(72) Routing Mode:

(73) Configuration device **28** may comprise a “routing mode”. In this mode, it connects individual registers **48** to one or more of the outputs **60**. The control instruments **26** may e.g. then use this for performing conditional execution of waveform generation during qubit state preparation of quantum processor **10**.

(74) In this mode, block definition **68** may e.g. comprise a list of registers and outputs to be interconnected.

(75) Boolean Mode:

(76) Configuration device **28** may comprise a “Boolean mode”. In this mode, it connects at least some of the registers **48** to at least some of the outputs **60** using Boolean operations, e.g. setting a given output to 1 only if two given registers have values of 1.

(77) In this mode, block definition **68** may e.g. comprise a Boolean description of the dependence of at least some of the outputs **60** on the values of at least some of the registers **48**.

(78) Optimization Mode:

(79) Configuration device **28** may comprise an “optimization mode”. In this mode, it programs at least part of the feedback block **54** (optionally with the help of coprocessor **67**) to form a numerical optimizer, i.e. a numerical minimizer or maximizer, for finding a minimum or maximum of a function depending on a plurality of values, in particular on at least some register values.

(80) The optimizer is advantageously adapted to find the minimum or maximum of a linear combination of a plurality of values. Such linear optimizers can be implemented using linear, non-iterative algorithms, which makes them fast and particularly suited to be implemented in feedback block **54**.

(81) In this mode, block definition **68** may e.g. comprise a description of the function to be minimized or maximized.

(82) The optimizer may e.g. be adapted to determine the most likely underlying error based on the readout data from ancilla qubits and/or to determine a specific control pulse that exhibits the highest gate fidelity and/or the readout pulse that exhibits the highest readout fidelity.

(83) Configuration device **28** may be adapted to generate structures, in particular logical structures, in the FPGA that comprise registers. Then, at least some of the data from the block definitions **68** may be directly routed into these registers, e.g. as a configuration dataset as mentioned above.

(84) Configuration device **28** may be adapted to structure feedback block **54** only. But may e.g. also be adapted to structure register bank **44**.

(85) It must be noted that, typically, configuration device **28** comprises at least two of the modes above, in particular at least three of them, in particular all of them. In particular, it advantageously comprises at least the LUT-mode and the routing mode and/or the LUT-mode and the FPGA direct programming mode and/or the LUT-mode and the ML-mode.

(86) The routing mode is particularly important, e.g. for the setup phase, and configuration device **28** therefore advantageously comprises at least the routing mode combined with at least one of the other modes.

(87) Further, it must be noted that FPGA configuration device **28** may be adapted to operate in

several modes when programming a single instance of system controller **18**. For example, it may form a lookup table in LUT-mode for connecting a first subset of the registers **48** with a first subset of the outputs **60**, in coprocessor programming mode for using coprocessor **67** to connect a second set of the registers **48** to a second sets of the outputs **60**, and/or in routing mode to interconnect some registers with some output values e.g. for state initialization.

(88) In more general terms, FPGA configuration device **28** may be adapted to program a first part of feedback block **54** in a first mode and a second part of feedback block **54** in a second mode different from the first mode.

(89) In one embodiment, configuration device **28** may be configured to merge the feedback block definition(s) **68** with e.g. VHDL definitions describing the circuitry of at least input interface **42**, register bank **44**, and output interface **61** in order to generate FPGA configuration data of at least part or the whole FPGA circuitry **19** of system controller **18**. For example, it may convert feedback block definition **68** to VHDL and then compile it, together with the VHDL definitions of the other components, whereupon it programs the FPGA **19** accordingly.

(90) Readout Instruments

(91) FIG. **5** shows an embodiment of a qubit readout instrument **24**. It comprises qubit interface circuitry **84** for interfacing with quantum processor **10**, e.g. comprising analog circuitry and at least one A/D converter. It further comprises a control section **86** as well as an interface section **88**.

(92) Interface section **88** is adapted to communicate through at least one of the input channels **20**, which includes a (logical or physical) sub-channel **20a** for the address data and a sub-channel **20b** for the measurement data.

(93) Each input channel **20** advantageously also comprises a system clock line **20c** with a master clock common to all readout instruments **24** and all control instruments **26** as well as a synchronization sub-channel **20d** providing a common time base to all readout instruments **24** and all control instruments **26**.

(94) Each input channel **20** advantageously also provides handshake functionality. In particular, it may carry a 'ready signal' indicative of the fact that the instrument is ready for new instructions.

(95) Control section **86** comprises at least one memory **86a** for storing at least one address of a register the measurement data is to be sent to. Memory **86a** may also store further operating instructions.

(96) Readout instrument **24** may also comprise a configuration input **90** connected to system configuration device **30**. In particular, readout instrument **24** may be adapted to receive instructions through configuration input **90** for changing the address and/or operating instructions in memory **86a**. This is described in more detail below.

(97) Advantageously, readout instrument **24** may be adapted to test one or more qubits from quantum processor **10** and, if they fulfill a certain condition, to write data into a specific register.

(98) Suitable programming can be entered into storage **86a** through configuration input **90**.

(99) Control Instruments

(100) FIG. **6** shows an embodiment of a qubit control instrument **26**. It comprises qubit interface circuitry **92** for interfacing with quantum processor **10**, e.g. comprising at least one D/A converter and analog circuitry. It further comprises a control section **94** as well as an interface section **96**.

(101) Interface section **96** is adapted to communicate through one of the output channels **22**, which includes a (logical or physical) sub-channel **22a** for the data from feedback block **54**.

(102) Each output channel **22** advantageously also comprises a system clock line **22b** with the master clock common to all readout instruments **24** and all control instruments **26** as well as a synchronization sub-channel **22c** providing a common time base to all readout instruments **24** and all control instruments **26**.

(103) Output channel **22** may e.g. be adapted to transmit multibit values to each control instrument **26**.

(104) Each output channel **22** advantageously also provides handshake functionality. In particular,

it may carry a 'ready signal' indicative of the fact that the instrument is ready for new instructions.

(105) Control section **94** comprises at least one memory **94a** for storing configuration data.

(106) Control instrument **26** also comprises a configuration input **98** connected to system configuration device **30** as described below. In particular, control instrument **26** may be adapted to receive instructions through configuration input **90** for changing its operating instructions.

(107) Advantageously, control instrument **26** may be adapted to test bits in the multibit value it received through its output channel **22** and to conditionally generate specific waveforms by means of interface circuitry **92**.

(108) Suitable programming can be entered into storage **94a** through configuration input **98**.

(109) System Configuration Device

(110) System configuration device **30** is adapted to program the readout instruments **24** and the control instruments **26**, i.e. to change the settings in their memories **86a** and **94a**.

(111) In the embodiment of FIG. **1**, system configuration device **30** is connected to the configuration inputs **90** and **98**. However, it may also communicate with the readout instruments **24** and the control instruments **26** through system controller **18**.

(112) Further, in the shown embodiment, system configuration device **30** is also connected to system controller **18** for controlling experiment controller **41** (see FIG. **2**).

(113) System configuration device **30** may comprise a system compiler **31** adapted to convert an (e.g. textual) experiment description into a configuration to be sent to the readout instruments **24**, control instruments **26**, and/or system controller **18**. The experiment description may e.g. specify at least one, in particular several, of the following experimental settings: pulse shapes to be generated by the control instruments **26**, pulse sequences to be generated by the control instruments, optionally as synchronized by experiment controller **41**, pulse times of the pulses generated by the control instruments **26**, readout times for the readout instruments **24**, register addresses for the register data from the readout instruments **24**.

(114) System configuration device **30** may further be adapted to send "experiment flow data" to synchronization device **40**, i.e. to experiment controller **41**. Experiment controller **41** is adapted to operate the control instruments **26** and the readout instruments **24** over a series of e.g. **1000** control and readout cycles of quantum processor **10** as a function of this experiment flow data.

(115) System configuration device **30** may then e.g. modify the experiment and then order experiment controller **41** to execute another series of control and readout cycles.

(116) System configuration device **30** is agnostic towards the specific configuration of feedback block **54**.

(117) Advantageously, though, system configuration device **30** is provided with at least "delay information" describing the delay between the input channels **20** and the output channels **22**, which depend on the configuration of feedback block **54**. This e.g. allows to synchronize the operations of the control instruments **26** with the ones of the readout instruments **24**, obviating the need of "wait" operations in the control instruments **26**.

(118) For this purpose, system configuration device **30** may be connected to FPGA configuration device **28** by means of a communication channel **100**.

(119) FPGA configuration device **28** can then calculate, in a feedback block parameter estimator **102** (cf. FIG. **3**), the delay information based on the currently selected feedback control definition(s) **68** and send this information to system configuration device **30**. Feedback block parameter estimator **102** can e.g. use timing analysis of FPGA **19** for calculating the delay information. Timing analysis software is e.g. provided as software packages by FPGA manufacturers for the FPGA devices they sell.

(120) Another item of information that may be useful for the operation of system configuration device **30** is a "sensitivity list", which describes which output **60** depends on which input **57**.

(121) In one example, system configuration device **30** may be adapted to receive an experiment

description for a given control instrument **26**. A (non-limiting) example of such an experiment description for a certain control instrument **26** may e.g. look like this: if (condition[4]) playWave(A); else play Wave(B);

(122) When compiling these instructions, configuration device **30** transmits programming to memory **94a** of the given control instrument, which, when being executed by the control instrument **26**, makes it test bit 4 of the value it received through its output channel **22**. If said bit is 1, it causes its interface circuitry **92** to play a first waveform. A, otherwise a second waveform B.

(123) In another illustrative example, system configuration device **30** may be adapted to receive an experiment description for a given readout instrument **24** e.g. as follows: measure(qubits={0,0,1,1,0}, reg={0,0,0x10,0x11, 0});

(124) When compiling these instructions, configuration device **30** transmits programming to memory **86a** of the given readout instrument **24**, which—when executed by the readout instrument—makes it read out qubits 2 and 3 of the available 5 qubits. It then sends the two binary readout results to addresses 0x10 and 0x11 through its readout channel **20** to system controller **18**, which causes the result to be written into the register with address **16** and **17** in register bank **44** and the respective new data signals are strobed.

(125) System Architecture

(126) Advantageously, the readout instruments **24** and the control instruments **26** are devices separate from system controller **18**. Advantageously, the channels **22** and **20** are serial channels for simpler connection.

(127) Also, FPGA configuration device **28** is advantageously a device separate from system controller **18**, e.g. a regular computer or dedicated hard- and software.

(128) Also, system configuration device **30** is advantageously a device separate from system controller **18** and also separate from the instruments **24**, **26**. It may e.g. also be a regular computer or dedicated hard- and software.

(129) FPGA configuration device **28** and system configuration device **30** may be separate devices or implemented as a common device.

(130) Advantageously, two devices are considered to be “separate” if they are connected by user-separable plug-in connectors only and/or if they are connected by a plug-in bus system, such as by a PXI backplane.

(131) Notes

(132) The term “system” as used herein denotes an apparatus with hardware as well as suitable software to be adapted and structured to carry out the functionality described.

(133) The present system allows the user to specify the feedback block definition(s) **68** only, without having to specify the configuration of the rest of the FPGA circuitry **19** implemented by system controller **18**, which greatly simplifies the task of configuring the system controller **18**.

(134) In other words, the FPGA configuration device **28** is advantageously adapted to receive information (through feedback block definition **68**) specifying the configuration of feedback block **54** only and then to configure the FPGA(s) **19** of system controller **18** to implement not only feedback block **54** but also at least input interface **42**, register bank **44**, and output interface **61**. It may e.g. do this by merging the feedback block definition(s) **68** with FPGA configuration data implementing the parts **42**, **44**, and **61** and then reprogramming the whole FPGA **19**, and/or by reprogramming the feedback block **54** of the FPGA(s) **19** while leaving the FPGA parts implementing the input interface **42**, register bank **44**, and output interface **61** unchanged.

(135) While there are shown and described presently preferred embodiments of the invention, it is to be distinctly understood that the invention is not limited thereto but may be otherwise variously embodied and practiced within the scope of the following claims.

Claims

1. A system for controlling a quantum processor comprising a system controller having a plurality of input channels and output channels, wherein each input channel carries measurement data, a plurality of qubit readout instruments connected to the input channels, a plurality of qubit control instruments connected to the output channels, wherein said system controller comprises at least one FPGA forming at least a) an input interface connected to said input channels, b) a register bank comprising a plurality of registers connected to said input interface, c) a programmable feedback block having inputs connected to said register bank and outputs, d) an output interface connecting the outputs of the programmable feedback block to the output channels, wherein the input interface is adapted to send the measurement data to the registers, wherein said system further comprises an FPGA configuration device connected to said at least one FPGA, wherein said FPGA configuration device is adapted to receive at least one feedback block definition, and program a section of said FPGA to form said feedback block in accordance with said feedback block definition.
2. The system of claim 1 wherein said FPGA configuration device has several modes of operation, wherein the different modes are adapted to program the feedback block to implement different algorithms for connecting the registers to the outputs of the feedback block.
3. The system of claim 2 wherein said FPGA configuration device has a LUT-mode, wherein, in said LUT-mode, said FPGA configuration device is adapted to program said section to form a lookup table storing a plurality of coefficients, wherein said lookup table comprises a selection input connected to said register bank and a result output connected to said output interface.
4. The system of claim 2 wherein said FPGA configuration device has an ML-mode, wherein, in said ML-mode, said FPGA configuration device is adapted to program said section to implement a machine learning model, and in particular wherein said machine learning model implements at least one of an artificial neural network, a decision tree, a support vector machine, a regression analyzer, a Bayesian network, and a genetic algorithm.
5. The system of claim 2, wherein said FPGA configuration device has an FPGA direct programming mode, wherein, in said FPGA direct programming mode, said FPGA configuration device is adapted to receive gate-level instructions for programming the section of the FPGA, and in particular wherein said FPGA configuration device is adapted to receive a set of VHDL instructions and to use said set for programming the gates in said section.
6. The system of claim 2 further comprising a coprocessor connected to said FPGA, wherein said FPGA configuration device has a coprocessor programming mode, wherein, in said coprocessor programming mode, said FPGA configuration device is adapted to program said section to exchange data with said coprocessor.
7. The system of claim 2, wherein said FPGA configuration device has a routing mode, wherein in said routing mode said FPGA configuration device is adapted to program said section to connect individual registers to one or more outputs of said feedback block.
8. The system of claim 2, wherein said FPGA configuration device has a Boolean mode, wherein in said Boolean mode said FPGA configuration device is adapted to program said section to connect at least some of the registers to at least some of the outputs-using Boolean operations.
9. The system of claim 2, wherein said FPGA configuration device has an optimization mode, wherein in said optimization mode said FPGA configuration device is adapted to program said section to implement a numerical optimizer, in particular a linear numerical optimizer.
10. The system of claim 2 wherein said FPGA configuration device is adapted to program a first part of the feedback block in a first mode and a second part of the feedback block in a second mode different from the first mode.
11. The system of claim 1 wherein said FPGA configuration device is adapted to merge said feedback block definition with information descriptive of at least the input interface, the register bank, and the output interface in order to generate FPGA configuration data, and program said at least one FPGA according to said FPGA configuration data in order to form said input interface,

said register bank, said feedback block, and said output interface in said at least one FPGA.

12. The system of claim 1 wherein each input channel carries address data in addition to said measurement data, said input interface comprises address decoder circuitry connected to said input channels, and the input interface is adapted to send the measurement data to one of the registers as specified by the address data.

13. The system of claim 1 further comprising a system configuration device connected to said readout instruments and said control instruments and adapted to program said readout instruments and said control instruments.

14. The system of claim 12 further comprising a system configuration device connected to said readout instruments and said control instruments and adapted to program said readout instruments and said control instruments, wherein each readout instrument comprises at least one memory storing an address to be transmitted as address data together with the measurement data on the input channel connecting the readout instrument to the system controller, and wherein said system configuration device is adapted to write the address into said address memory.

15. The system of claim 13 wherein said system configuration device comprises a delay input connected to said FPGA configuration device for receiving a value descriptive of a time delay between the input channels and the output channels in a current configuration of said system controller, and wherein said FPGA configuration device is adapted to calculate said time delay as a function of said feedback control definition.

16. The system of claim 1 wherein said system controller further comprises a synchronization device and wherein each input channel and output channel further carries a system clock line and a synchronization channel, wherein said synchronization device is adapted to synchronize said control instruments and said readout instruments by said system clock lines and said synchronization channels.

17. The system of claim 16 further comprising a system configuration device connected to said readout instruments and said control instruments and adapted to program said readout instruments and said control instruments, wherein said system configuration device is connected to the synchronization device and adapted to send experiment flow data to the synchronization device, and wherein the synchronization device is adapted to operate said control instruments and said readout instruments over a series of control and readout cycles of said quantum processor as a function of said experiment flow data.

18. The system of claim 1 wherein each register comprises a register data output and a new value output, both of which are connected to said feedback block, wherein said register data output carries a signal indicative of a value stored in the register, and said new value output carries a signal indicative of a write operation in the register.

19. The system of claim 18 wherein the feedback block is adapted to initiate a recalculation if at least some of the new value outputs flag a write operation.

20. The system of claim 19 wherein the feedback block comprises a sensitivity list storing a sensitivity information indicative of a subset of the registers, wherein the feedback block is adapted to initiate the recalculation if a given combination of the new value outputs of the registers of the subset flags a write operation.

21. The system of claim 20 wherein said FPGA configuration device is adapted to set the sensitivity information, in particular as a function of said feedback block definition.

22. The system of claim 1 wherein the FPGA configuration device is adapted to receive information specifying the configuration of the feedback block only and then to configure the at least one FPGA of the system controller to implement not only the feedback block but also the input interface, the register bank, and the output interface.

23. Use of the system of claim 1 for controlling a quantum processor.

24. A system for controlling a quantum processor comprising a system controller having a plurality of input channels and output channels, wherein each input channel carries measurement data, a

plurality of qubit readout instruments connected to the input channels, a plurality of qubit control instruments connected to the output channels, a system configuration device connected to said readout instruments and said control instruments and adapted to program said readout instruments and said control instruments, wherein said system controller comprises at least one FPGA forming at least a) an input interface connected to said input channels, b) a register bank comprising a plurality of registers connected to said input interface, c) a programmable feedback block having inputs connected to said register bank and outputs, d) an output interface connecting the outputs of the programmable feedback block to the output channels, wherein the input interface is adapted to send the measurement data the registers, wherein said system further comprises an FPGA configuration device connected to said at least one FPGA, wherein said FPGA configuration device is adapted to receive at least one feedback block definition, and program a section of said FPGA to form said feedback block in accordance with said feedback block definition, wherein said system configuration device comprises a delay input connected to said FPGA configuration device for receiving a value descriptive of a time delay between the input channels and the output channels in a current configuration of said system controller, and wherein said FPGA configuration device is adapted to calculate said time delay as a function of said feedback control definition.

25. A system for controlling a quantum processor comprising a system controller having a plurality of input channels and output channels, wherein each input channel carries measurement data, a plurality of qubit readout instruments connected to the input channels, a plurality of qubit control instruments connected to the output channels, wherein said system controller comprises at least one FPGA forming at least a) an input interface connected to said input channels, b) a register bank comprising a plurality of registers connected to said input interface, c) a programmable feedback block having inputs connected to said register bank and outputs, d) an output interface connecting the outputs of the programmable feedback block to the output channels, wherein the input interface is adapted to send the measurement data the registers, wherein said system further comprises an FPGA configuration device connected to said at least one FPGA, wherein said FPGA configuration device is adapted to receive at least one feedback block definition, and program a section of said FPGA to form said feedback block in accordance with said feedback block definition, wherein each register comprises a register data output and a new value output, both of which are connected to said feedback block, wherein said register data output carries a signal indicative of a value stored in the register, and said new value output carries a signal indicative of a write operation in the register.

26. The system of claim 25 wherein the feedback block comprises a sensitivity list storing a sensitivity information indicative of a subset of the registers and wherein the feedback block is adapted to initiate the recalculation if a given combination of the new value outputs of the registers of the subset flags a write operation.
